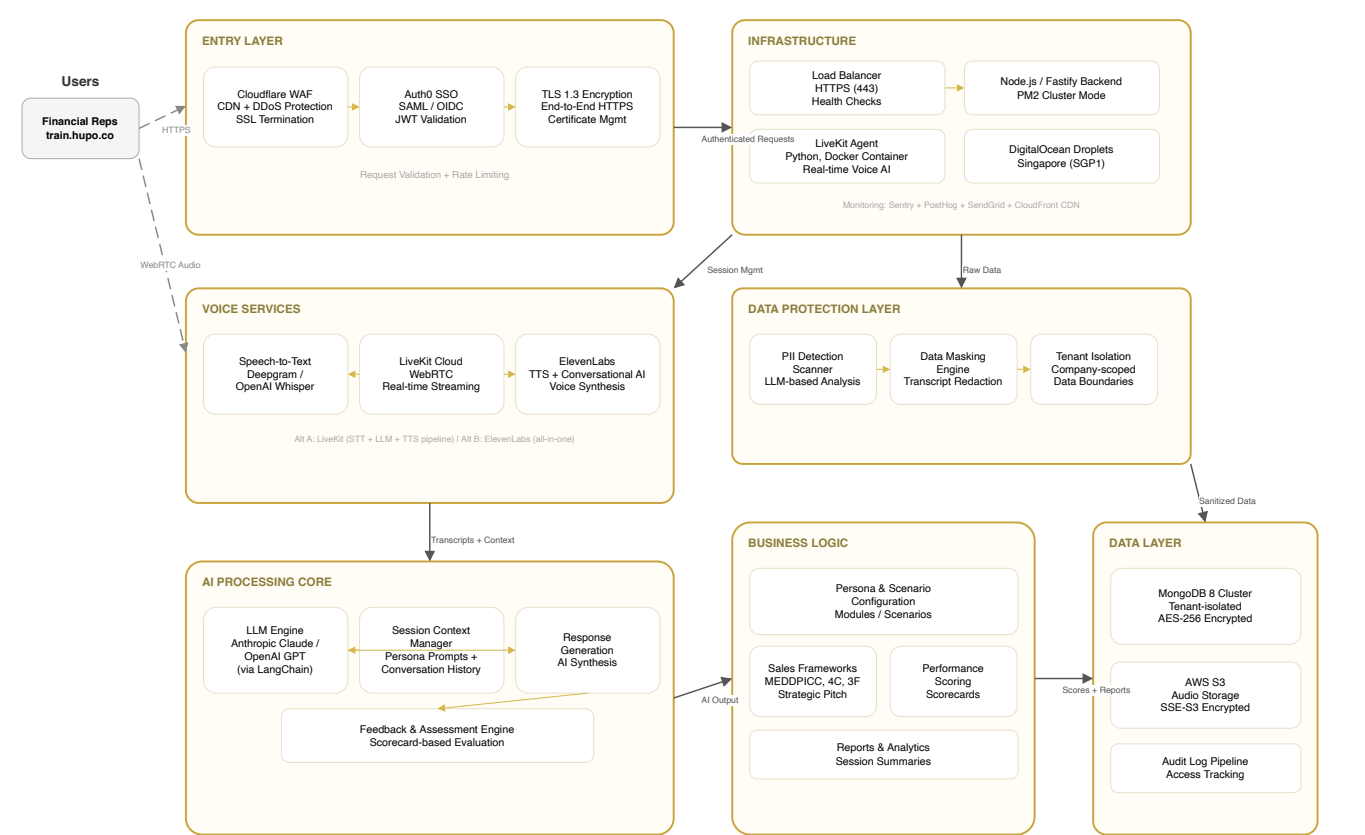


High-Level Architecture Diagram

For GE questionnaire — Section 1: High-Level Hosting & Integration Posture



Summary

Component	Provider	Region	Notes
Frontend (SPA)	Cloudflare Pages	Global CDN	React, browser-only
CDN / WAF	Cloudflare	Global	DDoS protection, SSL termination
Authentication	Auth0 (Global SaaS)	US	SAML / OIDC, JWT, multi-tenant SSO
Backend API	DigitalOcean	Singapore (SGP1)	Node.js / Fastify, PM2 cluster
Database	DigitalOcean Managed	Singapore (SGP1)	MongoDB 8, Primary + 2 Standbys, AES-256
Voice AI Agent	DigitalOcean	Singapore (SGP1)	Python, Docker container
Voice Server	LiveKit Cloud (Global SaaS)	Multi-region	WebRTC real-time streaming
Audio Storage	AWS S3	ap-southeast-1	SSE-S3 encryption
Media CDN	AWS CloudFront	Global	Static assets

AI/LLM	Anthropic / OpenAI	US	Via LangChain
TTS / Conversational AI	ElevenLabs	Global	Voice synthesis + direct mode
STT	Deepgram / OpenAI Whisper	Global	Speech-to-text
Error Tracking	Sentry	Global SaaS	
Analytics	PostHog	Global SaaS	
Email	SendGrid	Global SaaS	Transactional email

Architecture Layers

Entry Layer

All user traffic enters through Cloudflare (CDN, WAF, DDoS protection, SSL termination), then authenticates via Auth0 (supports SAML for GE IdP integration). All communication is TLS 1.3 encrypted end-to-end.

Infrastructure

Application runs on DigitalOcean Singapore (SGP1) with a managed load balancer. The Fastify backend runs in PM2 cluster mode. A separate LiveKit Agent (Python, Docker) handles real-time voice AI processing.

Voice Services

Two voice provider paths: **LiveKit** (STT → LLM → TTS pipeline for select languages) and **ElevenLabs Conversational AI** (all-in-one default). Speech-to-text via Deepgram / OpenAI Whisper. Text-to-speech via ElevenLabs.

Data Protection Layer

PII detection using LLM-based scanning before data storage. Data masking engine redacts sensitive information from transcripts. All data is tenant-isolated with company-scoped boundaries.

AI Processing Core

LLM engine (Anthropic Claude / OpenAI GPT via LangChain) handles response generation during roleplay sessions. Session context manager maintains persona prompts and conversation history. Feedback and assessment engine provides scorecard-based evaluation.

Business Logic

Configurable personas, scenarios, and sales training modules (MEDDPICC, 4C, 3F, Strategic Pitch, etc.). Performance scoring with detailed scorecards. Reports and session analytics.

Data Layer

MongoDB 8 cluster with tenant isolation (company-scoped), encrypted at rest (AES-256). Audio files stored in AWS S3 with SSE-S3 encryption. Audit log pipeline for access tracking.